



Future

Preliminary Design Review

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Equipe Rocket UTFPR

2019 COBRUF Aerospace Competitions



About Equipe Rocket		
Target Segments Aerospace, aeronautics, avionics and telemetry, missile and rocket development industries. University aerospace studies. National security.		Value Propositions The CFD and structural analyses made on the engine ensure we have a quasi one dimensional flow insine the engine, and prevent the possibility of a backflow in the nozzle throat. The development of a simple passive recovery system prevents failure in recovery and helps to increase to security of operation and reusability of our rockets. Our avionics systems with acceleration sensors, gps and high speed data transmitter gives us security to find the rocket even with a high dispersion zone of recovery.
Relationships The development of our project helps improve students learning process and gives extra credits for the hours worked.	Channels Social Medias, Radio, congress exhibitions.	
Resources Composite materials laboratory, machining shop, CNC laboratory, chemistry laboratory, Ansys license, solidworks license, Autocad license, Matlab license		
Revenue Streams What were the main revenue sources of your project? (Ex.: Sponsors, University Add, Raffles, etc).	Cost Structure Raw Materials, 3d printing, and manufacturing of tools and equipment's	Activities First we design a 3d model of the rocket to evaluate the total mass and center of mass after that the engine was projected to have enough impulse to reach the apogee, than the airframe was designed to have a good center of pressure and decrease the auto stabilization velocity and to stand the hydrodynamic pressure and we build a simple recovery system to have more confiability. All project was designed in Solidworks and the structural analyses on Ansys
Partners UTFPR JH Construção Ferbomat		



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List of Symbols:

A	amplitude of oscillation
a	cylinder diameter
C_p	Pressure coefficient
C_x	force coefficient in the x direction
C_y	force coefficient in the y direction
c	Chord
dt	Time step
F_x	X component of the resultant pressure force acting on the vehicle
F_y	Y component of the resultant pressure force acting on the vehicle
I_t	Total impulse
M_p	Specific mass of the propellant
I_{sp}	Specific impulse
G_0	Earth Gravitational acceleration
F	Total force
V	Velocity
D	External diameter
d	Internal diameter
L	length
R	Radius
T_n	Normalization time
T_{ign}	Burn time
T_{ac}	Inhibitor wall thickness



1. Detailed Theoretical Design Description

1.1. Propulsion

The engine has the function of propelling the rocket to the desired altitude. The subsystem is responsible for analyzing the choice of materials for the various engine parts, taking into account the stresses and high temperatures involved in combustion. Still, it is the subsystem's responsibility to choose and manufacture the propellant, the fuel used to generate thrust.

The engine design for a 6000 ft launch was performed considering optimal design parameters such as combustion analysis, weight of onboard equipment, cost of materials and methods available for manufacturing.

Considering the total rocket mass and target altitude data, the projected thrust is 2200 N.s. From the design parameters were made combustion simulations seeking progressive burning profile and 1800N of max thrust.

A project flowchart was generated and for each step of the engine development process reports were prepared and used solution comparison tools.

Projeto de Motor Foguete Sólido

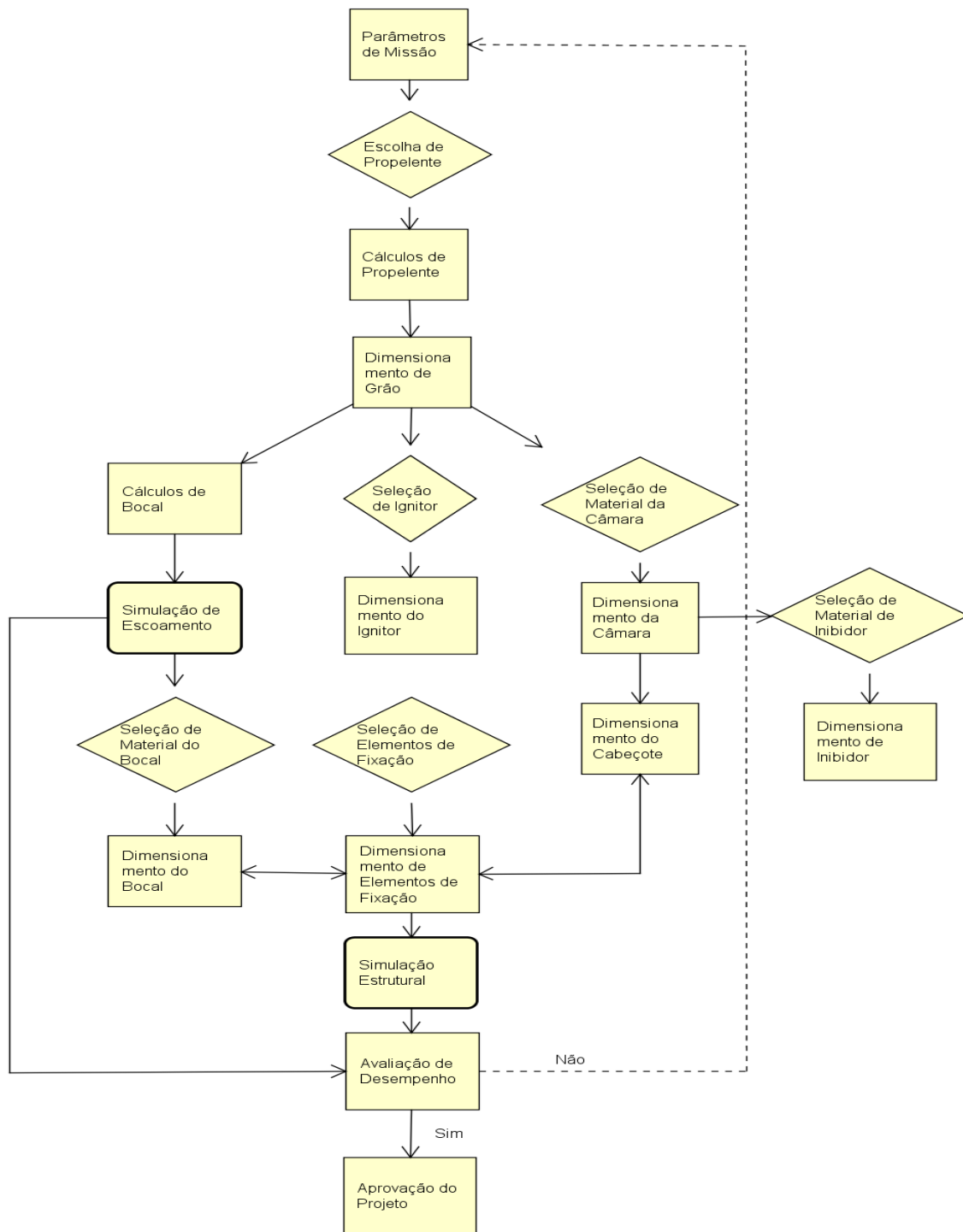


Figure 1 - Flowchart of propulsion development

1.1.1. Propellant

Propellant is one of the most important parts of a rocket engine's design, since much of what is decided takes a predetermined course. In this case, we chose to use solid type fuels for their simplicity of manufacture and control in the result and stability calculations.

Decision Parameters

Pugh's table method was used, which consists of choosing a material as the basis for grades on a scale of -3 to +3 of other materials in relation to each evaluation criterion, where -3 is much lower than the base material. chosen, +3 is much higher and 0 equals a predetermined criterion. The Pugh matrix is a simple and efficient way to add weights to your chosen criteria, making the choices you make from it even more reliable for whatever purposes.

RELATÓRIO DE DECISÃO PROPELENTES DO TIPO SÓLIDO EM MISTURA DE COMPONENTE		DESIGN CONCEPTS		
Criteria/Concept	Weigth	baseline made	concept made	concept made
		Sucrose	Dextrose	Sorbitol
		Baseline	Concept 1	Concept 2
Custo	3	0	-2	-1
Facilidade de Fabricação	2	0	+1	+3
Segurança	2	0	0	+2
Impulso Específico	2	0	+1	-1
Estocagem	1	0	+1	+1
Curva de Empuxo	4	0	0	-2
Temperatura de Combustão	1	0	0	+1
Total S +		0	3	7
Total S -		0	2	4
Total S 0		0	0	0
Total with weigth		0	-1	-1
Rank with weigth		1	2	2

Figure 2 - Propellant pugh matrix

The greater weight (4) was for the thrust curve, such a more normalized profile curve will facilitate future calculations that are of great importance. Cost was 3 because one of the team's current needs is to spend the least amount of money per application on each project. Safety, ease of manufacture, and specific thrust are weight 2 because from special measures that can be taken.

Propellant: KNSu was selected due to its energy density, manufacturing safety and operation. The oxidant-propellant rate is 65% -35% respectively.



Figure 3 - Knsu grain

Such criteria will have very small differences. Stocking is weight 1 because it is not the intention of the team at this time, and in the case of combustion temperature (weight 1) it may be a minor criterion if the ignitor and inhibitor are of higher / more efficient qualities (which is not directly linked to their cost).

It was agreed among the propulsion subsystem members, and based on the evaluation method adopted, that the best propellant for the current condition of the team should be the mixture of Potassium Nitrate and Sucrose.

1.1.2. Chamber Material Selection

To make the proper choice of the material of the combustion chamber, it is necessary to know some parameters such as density, mechanical and thermal resistance of the material, as these values influence its mechanical properties, such as modulus of elasticity, yield strength, yield strength. thermal expansion, thermal conductivity [1].

This section is intended to show the data and analysis used by the propulsion subsystem for the choice of combustion chamber material. To determine the material chosen, a

series of analyzes using the PUGH Matrix methodology, as well as the aid of the casing program [2] developed by Nakka for Excel.

Project Parameters

The PUGH matrix was used to assist in choosing the material to be used in the combustion chamber. As with the parameters mentioned above, the cost of the chosen material must be taken into account. This parameter is critical in the analysis of the material, and material that is appropriate for the project and that is in line with the Team budget should be defined. Still, the manufacture of the chamber with the material chosen should be carefully analyzed, as this is the first time the team will manufacture the engine, having no experience in handling the material.

In order to select possible materials for the project, we used the “Material Properties” table provided by Nakka in the “casing” program. Among the various materials available, low carbon steel, alloy steel, stainless steel and aluminum were selected as possible materials for choosing the project material.

To analyze the cost of the pre-selected materials a quick search was made and found the following order:

- Low carbon steel (SAE 1020);
- Low alloy steel (A116);
- Stainless steel (SAE 301);
- Aluminum.

Where SAE 1020 steel was the cheapest and the most expensive aluminum.

To analyze the weight of each material used the density as a selection parameter. Therefore, Table 1 reports some properties of the pre-selected materials taken from MatWeb [3].

Tabel 1 Material properties

Material	Densidade [g/cm ³]	σ_e [MPa]	σ_t [MPa]	Condutividade térmica [W/mK]
Aço SAE 1020	7.87	330	450	51.9
Aço SAE 4130	7.85	435	670	30.1
Aço SAE 301	8.03	205	515	16.3
Alumínio 6061 - T6	2.7	276	310	167

Autor: Autoria própria.

In addition to the data obtained in Table 1, it is possible to analyze the behavior of the pre-selected materials in Figures 1 and 2, where SAE 301 stainless steel and SAE 4130 low alloy showed the lowest ratio between yield strength loss by SAE 4130 and SAE 1020 low alloy steels showed lower ratio between loss of temperature resistance limit in Figure 2.

Furthermore, it is known that the temperature in the combustion chamber can reach high values, requiring an inhibitor and a material to support this design requirement. However, you should not forget the cost, weight and manufacture of the combustion chamber in choosing the material chosen.

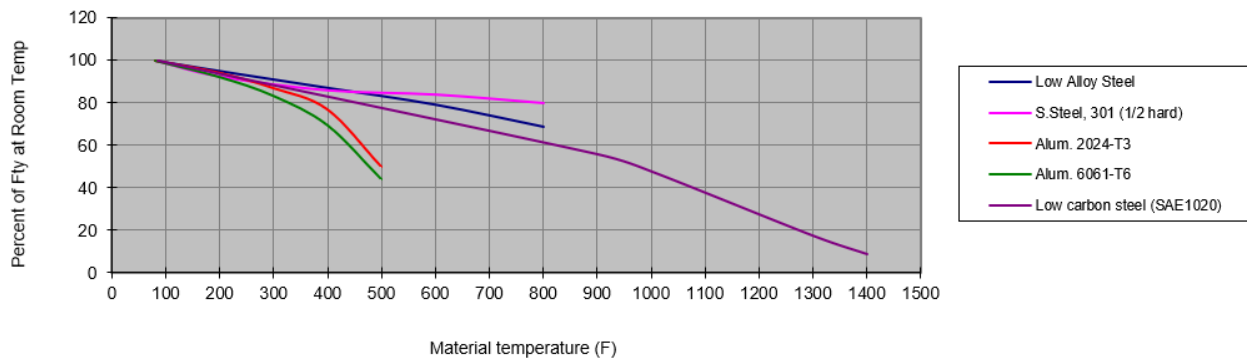


Figure 4 - Effect of temperature on yield stress (Fty) after half an hour of exposure.

Source: R. A. Nakka.

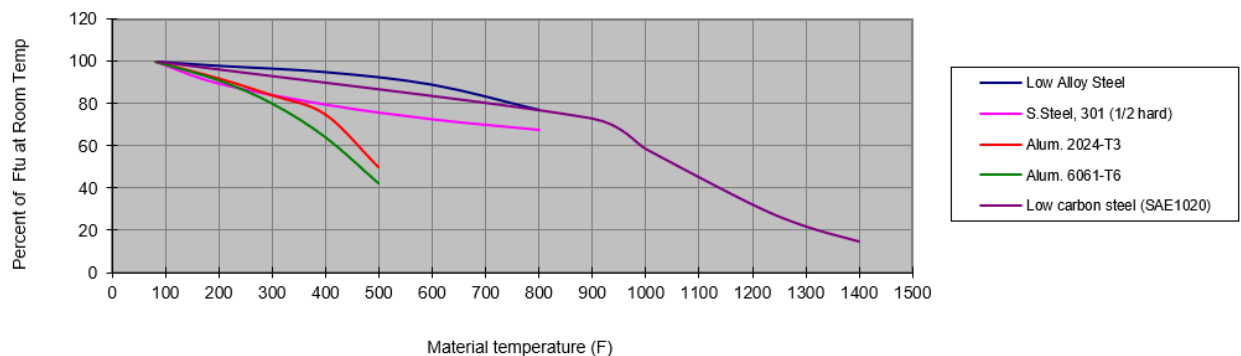


Figure 5 - Effect of temperature on resistance limit (Ftu) after half an hour of exposure.

Source: R. A. Nakka.

Thus, with the mentioned parameters and possible pre-selected materials, the PUGH Matrix was assembled as observed in Figure 6. Thus, analyzing Figures 4 and 5, it is observed that the best material was SAE 4130 alloy steel due to its mechanical properties and, mainly, its good behavior to temperature variation.

PUGH CONCEPT SELECTION					
<i>Material da Câmara</i>		DESIGN CONCEPTS			
		baseline image	concept image	concept image	concept image
		Aço Baixo Carbono	A116	Aço Inox	Alumínio
Criteria/Concept	Weigth	Baseline	Concept 1	Concept 2	Concept 3
Custo	2	0	-1	-2	-2
Peso	2	0	0	-1	+2
Resistência Mecânica	3	0	+2	+1	-1
Resistência Térmica	3	0	+1	+1	-2
Fabricação	1	0	-1	-2	+1
Total $\Sigma +$		0	3	2	3
Total $\Sigma -$		0	2	5	5
Total $\Sigma 0$		0	0	0	0
Total with weigth		0	6	-2	-8
Rank with weigth		2	1	3	4

Figure 6 - PUGH for chamber material

According to the data obtained from the analysis of the pre-selected materials and the PUGH Matrix, the Rocket Team UTFPR-CP propulsion subsystem concluded that the engine chamber will be A116 alloy steel, as its mechanical properties, such as mechanical strength and thermal, and its weight in relation to the others proved to be more adequate for the preparation of the proposed project. Still, its cost compared to SAE 301 stainless steel or 6061-T6 aluminum is more suited to the Team's budget.

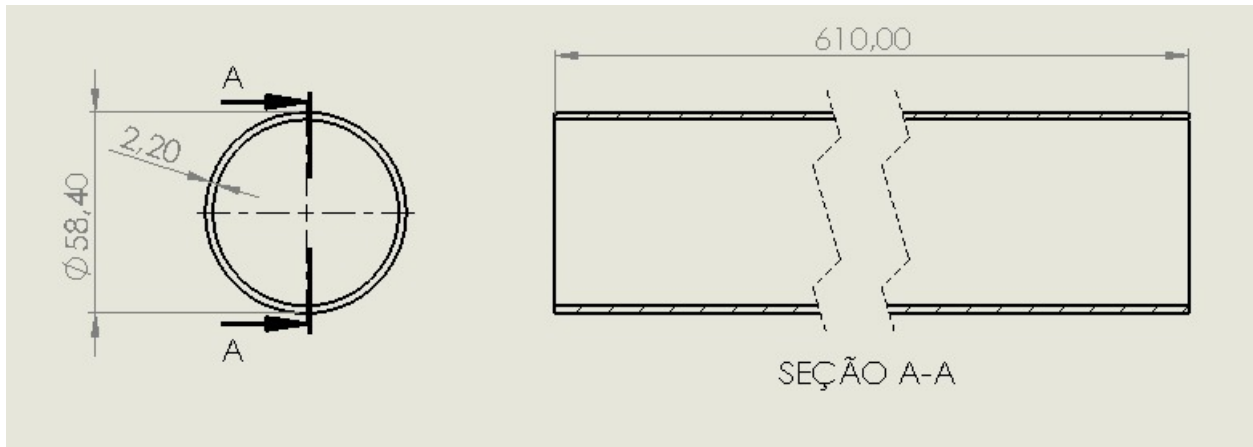


Figura 7 - Combustion chamber

1.1.3. Nozzle Material Selection

The nozzle can be described as one of the fundamental parts in the construction of a rocket, since it has the function of channeling and accelerating the combustion products produced by the propellant burning in order to maximize the output speed of these products.

The choice of material for the nozzle is of equal importance as it must withstand high temperatures and pressures at the time of firing. Thus, a material capable of withstanding such conditions is necessary so that no failure occurs.

Project Parameters

The choice of materials followed took into account various design factors, from cost, handling, ease of obtaining, mechanical strength, thermal resistance to their specific weight.

After a quick search, we can select some of the most interesting materials for our use, they are:

- SAE 1018 steel

1018 steel is a carbon steel with approximately 0.18% carbon. This steel has excellent weldability. This carbon steel has a chemical composition that allows a good balance between its mechanical properties of strength, toughness and ductility. Machinability and hardness increase when SAE 1018 is hot rolled.

Density: 7.78 g / cm³

Flow Limit: 370 Mpa

Tensile Strength: 440 Mpa

Elasticity Module: 205 Gpa

- SAE 8630 steel

SAE 8630 steel has chromium, nickel and molybdenum in its alloy (chemical composition). It has mechanical properties suitable for in various areas.

Density: 7,85 g / cm³

Flow Limit: 550 Mpa

Tensile Strength: 620 Mpa

Modulus of elasticity: 190-210 Gpa

- SAE 12L14 Steel

It is characterized by the controlled sulfur content (S) in its chemical composition, which facilitates the cutting of the material during the machining process.

Density: 7,9 g / cm³

Flow Limit: 230 Mpa

Tensile Strength: 390 Mpa

- Graphite

Several high performance rocket nozzles are made of graphite. Graphite is an excellent material for nozzles for many reasons, we know that most of the graphite used in everyday life is very soft and therefore we cannot use it to make nozzles. However, we are talking about graphite with a more crystallized structure, with a specific density of the order of 1.6-2.1, where even in situations of high stress it still withstands high temperatures (mp 3550 ° C). Matrix puhg material nozzle and headstock

PUGH CONCEPT SELECTION

<i>Material do Bocal</i>		DESIGN CONCEPTS			
		baseline image	concept image	concept image	concept image
		1018	Grafite	8630	12L14
Criteria/Concept	Weigth	Baseline	Concept 1	Concept 2	Concept 3
Custo	3	0	-3	-1	-2
Resistência Térmica	2	0	+3	+1	0
Fabricação	3	0	-2	0	0
Resistência Mecânica	2	0	-3	+1	-1
Total $\Sigma +$		0	3	2	0
Total $\Sigma -$		0	8	1	3
Total $\Sigma 0$		0	0	0	0
Total with weigth		0	-15	1	-8
Rank with weigth		2	4	1	3

Figure 8 Pugh Nozzle and bulkhead

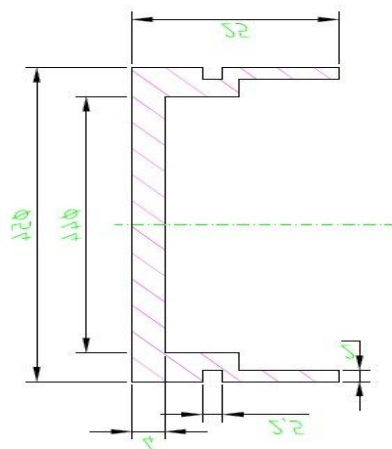


Figura 9 - Bulkhead project

1.2. Avionics

A vehicle avionics is defined as the entire onboard electronics functions, including the telemetry, processing and data transmission, cargo control and the on-board computer. Where each system has its specific function.

All avionics systems are modular and usually have redundancies, all electronic system embedded in the rocket is essential for acquisition, treatment and flight data storage, the rocket path being uncontrolled.

Subsystems are responsible for specific tasks such as telemetry that does all acquisition of flight data such as pressure, acceleration and data engine health, the on-board computer that does all data processing and send them for data transmission and store all data on an SD card.

All data obtained during the flight are intended for further study, it is evaluated and compared with the expected conditions through the calculations and simulations. And using flight analysis and experiments, we developed a design that corrects past failures and to predict future failures.

We can verify the work performed by the avionics team in figure below, where we can see how the system is modular, the shape of the PCB and the layout of some components that make up the system.

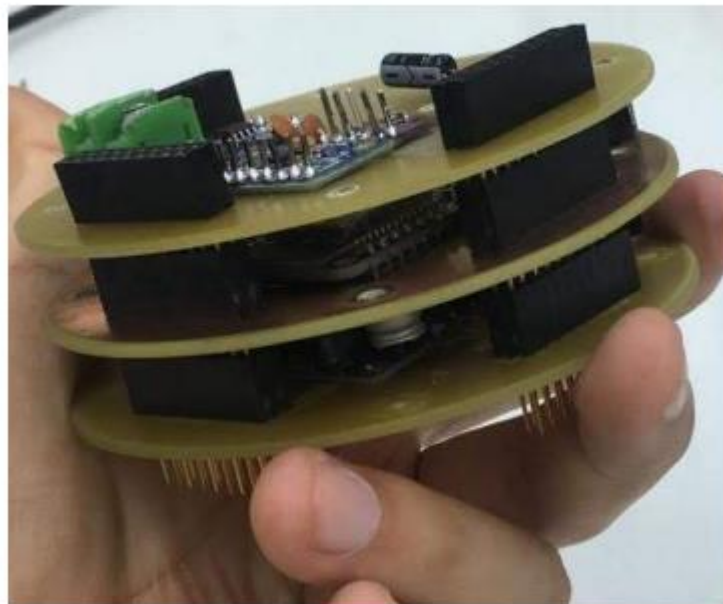


Figure 10 - Avionics Circuit

1.3. Structures

The fins and cone are responsible for the stability of the rocket in the air, important for guiding our rocket in the direction we launched it directly interferes with the safety of the launch. Choosing a suitable material ensures that our rocket has a Successful launch in the given direction.

Centralizers are responsible for centralizing any component and to kill them fixed in a certain structure. Normally one-piece rings are used predefined material, depending on the purpose of its use.

The main function of the centralizer is to maintain the rocket motor centered and fixed within the structure, thus allowing weight to be distributed evenly, making it easier to find the center of mass.



Figura 11 - Engine centralizers

1.3.1. Frame

The frame is one of the most important parts of the project, as it is the main structure, responsible for interfacing the internal and external environments. Ensuring a good choice of material is paramount to validating a successful launch of our rocket. From the Pugh matrix, considering our decision parameters, we chose the PVC (Polyvinyl Chloride) material, as it is a low cost and easily accessible material, a priority to our manufacturing reality, as we showed from the Pugh matrix.

1.4. Recovery System

1.4.1. System operation

The type of opening system chosen was the expulsion of the parachute by means of an elastic force provided by an air bag, with the aid of a drug, where the device responsible for unlocking the compartment door is a lock pin. , driven by a servo motor.

After accommodating the chute and the drogue inside the chamber and closing the door, we pressurize the air bag using a external pump.

The main factor in choosing this system was to maintain functionality without excessive complexity in order to simplify the device as much as possible to optimize openness, thus reducing the chances of system failure.

The servo motor will be driven at the moment of opening by means of a control center, later, through a rod connected to the locking pin, it will rotate its fixing point of the rod, generating a tensioning force, which will be responsible for pulling the locking pin along the rail (fixed to the door), making the parachute cover open.

This servo motor, was chosen because it allows, through third devices, a long distance control and good accuracy at opening.

The lock pin, which is responsible for keeping the door locked during the initial rocket course, was taken as an element to be applied, thanks to its simplicity of operation, operating in good conjunction with the servo motor. It has a good fastening, preventing the compartment door to be opened early, but also easy to run on the rail, preventing the pin from getting stuck during opening and the door not opening.

In addition, it has reduced cost and commercial ease, being an easily accessible product. Therefore, making its replacement for the continuation of the project easy and inexpensive.

The principle of operation of a parachute is basically to increase the area of an object thus obtaining a system with a large area of contact with the air, thus limiting the terminal velocity of fall of this object due to the deceleration promoted by the force. of drag.

For this, it is of utmost importance to choose a suitable parachute shape that provides the highest efficiency during rocket deceleration. In this way, a good parachute system will allow the projectile to fall to the ground without causing major damage to its own structure. Also, as one of the requirements of the competition, besides launching, is recovery, a proper parachute system is required.



1.4.2. The ejection system

The ejection system is of great importance to the operation of rocket recovery, as it has the function of expelling the parachute so that it can decelerate the rocket and leave it with a relatively low constant speed, so that the rocket as a whole is recovered undamaged to the point that it can be released again.

The choice of the model and the materials to be used is of great importance within the recovery system, as it must be able to eject the rocket's parachute and thus be able to recover it. the drag exerted on the fuselage of the rocket. Thus system must overcome all these barriers and cannot fail.

For the choice of system ejection type, several design factors were taken into account, such as cost, ease of obtaining, ease of handling, mechanical strength, risk of failure to buoyancy on landing. Initially we worked with two possibilities, spring drive and tube drive. Both start from the same idea, and are described below.

In the spring option, the parachute would be placed on a support attached to two coil springs and then pressed so that the door could be closed, with this procedure a force would be exerted against the cover.

In the air chamber option, the parachute would be placed over the air chamber and then the door closed, after which the air chamber would be inflated exerting a force against the cover.

For both options, at the time of system activation, a servo motor will pull a lock attached to the rocket body that will release the cover and with wind pressure it will be possible to release the parachute so that it can slow down and make the landing. rocket at a constant speed.

After evaluating the possibilities and analyzing their proper functioning, it was decided by the recovery subsystem which system would be chosen via the air chamber due to its ease of obtaining, handling and application, being therefore the most viable option at the moment.

1.4.3. Suspension Lines evaluation

The material chosen for use is paracord (also known as parachute rigging), which is very light, takes up little space and is very durable.

Paracord is a rigging constructed from a synthetic fiber called high-tech polyamide nylon, which allows for excellent strength and good elastic potential. Nylon PA is a semi-crystalline thermoplastic with low density and high thermal stability. Polyamides are among the most important and useful technical thermoplastics due to their excellent wear resistance, good coefficient of friction and very good temperature and impact properties. This excellent balance of properties makes nylon an ideal material for metal replacement in applications whose design requirements include high strength, toughness and weight reduction.

The paracord used is the type 3, more commonly called paracord 550. It has an approximate diameter of 4mm and has 7 braids inside, holding approximately 249kg load. Excellent value for money.

The choice of this material is due to its use in parachute, and also because it is an admirable equipment in terms of comparison between weight, volume and resistance.

1.4.4. Parachute Material choice

The parachute material chosen is the Rip Stop nylon 600, which is commonly used in military and purpose uniforms.

sports, such as firefighters, police, army, etc., so that they have greater durability. The main feature of Rip Stop fabric is its material made of nylon interlaced with another type of fabric such as polyester, polyamide, cotton etc, creating a locking surface. The application of chemical residues such as acrylic resin, silicone and PVC makes the material waterproof, further increasing its durability. Rip Stop is present in the production of parachute fabrics, balloons, camping tents and sleeping bags. Activities that use equipment that require effectiveness. If a hole happens to open a hole in the fabric, the interweaving of the threads prevents the hole from tearing even further. The cost-effectiveness of the material is excellent for its durability. The choice of this material is due to its combination of lightness, strength, cost-benefit, and also being commonly used in parachute fabrics. Rip Stop is considered the ideal fabric for equipment that is exposed to high wear or even exposure activities.



1.4.5. Servo motor

The servo motor is one of the main parts of the recovery, as it is used to pull the pin that will open the door and thus pull out the parachute.

It is very important to choose a suitable servo motor, as it needs to have a high torque to be able to turn and pull the pin, because if it is one with a low torque, it can end up breaking the servo and will not trigger the parachute, so for the rocket recovery, one must choose well, as no failure can occur.

1.4.6. Drogue

In our recovery system a kicker will be used to assist the main parachute, it is nothing more than a second smaller parachute that is pushed out at the time of actuation, before the main parachute, in addition this will be stuck to when released, it will also have the function of assisting in this activation.

For making the drug, we will use the same materials as the parachute, and in addition we will choose a simpler format, as it needs to be easily expelled from the compartment, and this will not have to have the same performance as the main, and also for ease of use. modeling as well as having little stitching, which reduces weight and catastrophic failures.

For these reasons, the format chosen was the square, which is basically a square fabric attached by 4 suspension ropes, which are attached to its 4 vertices, this kick drug has the advantages already mentioned over other formats, such as the semielliptic, and also It has simpler features than the X parachute. The size of this second parachute will depend mainly on the size of the opening door, as we have in mind to make it with a dimension easily pass through it, and furthermore it does not need to have a force exact for a good operation, for this reason it will be approximately 30x30cm.

2. Theoretical Design Methodology

2.1. Propulsion Design

2.1.1. Propellant-related sizing and calculations

Propellant grain refers to the geometric shape of the propellant when loaded inside a rocket and is responsible for various pressure and thrust characteristics throughout engine operation, as well as its burning time and the amount of propellant required. to the desired performance. This section aims to show the calculations, simulations and data used for the propellant grain sizing, as well as the assumptions used by the propulsion subsystem for its elaboration. To determine the grain type and the number of segments, a series of calculations were performed, as well as simulations with the aid of the SRM_2014 Excel program.

2.1.2. Project Parameters

The first step in the development of these activities is the definition of the design parameters. From the initial work, a mixture of Potassium Nitrate and Sucrose (65/35) was chosen as the propellant for the engine, the properties of this propellant composition. are given for a maximum operating pressure of 1000 psi (6894000 Pascal). This pressure is used as a reference for performance and propellant calculations and is often used in engine designs, so is also the pressure adopted in engine development. Using ProPEP3 software and KNSU properties table [1], the following values of interest were obtained for this work.

- Density: 1800 kg / m³
- Specific Impulse: 134 s

- Burn Rate: 0.0153 m / s
- Speed Characteristic: 895 m/s

Having also set the inside diameter of the rocket body to 70 mm, is a parameter of It is a common design that the combustion chamber diameter is 75% of this body diameter [2], so we have a chamber diameter of 0.0525 m. Regarding the desired performance of the engine, defining a class K engine with 1621 Newtons of medium thrust, will also define a burn time of approximately 1,42 s.

Hint! To directly convert inches to mm, simply type =25.4* number where number is dimension in inches.

Motor chamber:

Dc mm Chamber diameter (inside)
 Lc mm Chamber length (inside)
 Vc mm³ Chamber volume (empty)

Propellant grain:

Propellant type [See Note \[1\] below](#)

Do mm Outer diameter (initial)
 do mm Core diameter (initial)
 Lo mm Segment length (initial)
 N Number of segments

Outer surface: [select](#)
 Core surface: [select](#)
 Ends surface: [select](#)

Lgo mm Grain length (initial)
 Vg mm³ Grain volume (initial)
 Vt Volumetric loading fraction

p' grain g/cm³ Grain ideal density
 Density ratio (actual/ideal)

p grain g/cm³ Grain actual density
 m grain kg Grain mass (initial)

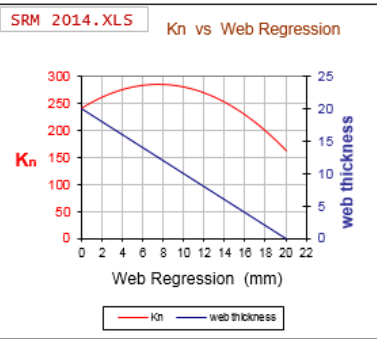
Abco mm² End burning area (initial)
 Abco mm² Core burning area (initial)
 Abso mm² Outer surface burning area (initial)
 Abo mm² Total burning area (initial)

Target MEOP: [select](#) Maximum chamber pressure (target)

Kn max: Ratio of Burning area / throat area (max)

Nozzle: Ato mm² Throat cross-section area (initial)

Kn vs Web Regression



Click to solve (1)

Kn max: 285
 Kn min: 163

Figure 12 - Grain dimensions

We decided to use a 8 grain setup with a core rod diameter of 10 mm and length of 65mm, the core surface and ends surfaces and exposed with a space between than of 3,75 mm. this give us the total mass of propellant of about 1,889 Kg.

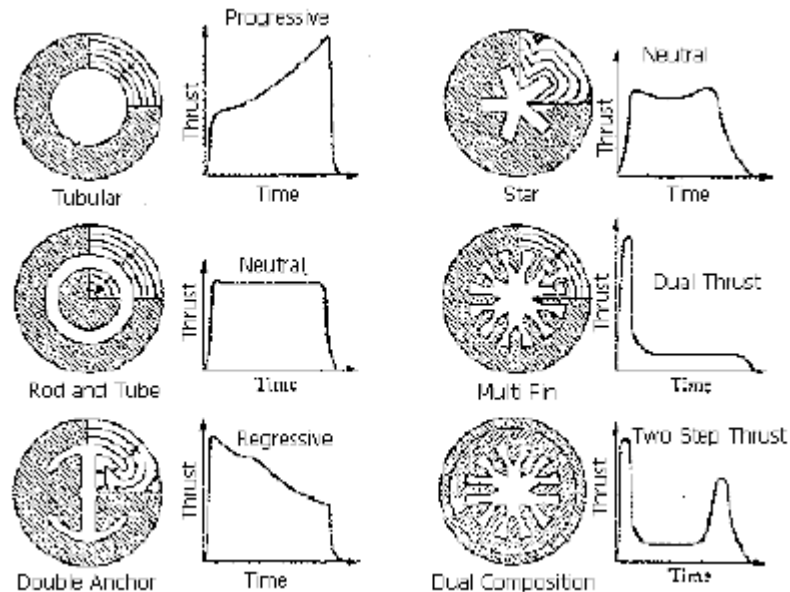


Figura 13 - Grain form

The image above shows the effects that different grain hole geometries have on the thrust curve, in the case of this design a neutral burn was chosen to achieve greater combustion efficiency and avoid thrust irregularities that could damage more sensitive vehicle components, like avionics systems. To obtain this thrust curve would require a star-shaped hole or using a circular channel, both geometries present difficulties from the manufacturing point of view, however there is another alternative, grain bates, which has a circular hole and multiple spaced grains, thus increasing their burning area and creating a neutral thrust curve [3], so this was the geometry chosen for this project.

2.1.3. Grain Sizing

The sizing process begins by determining the propellant mass required to meet our requirements using the following formula.

$$It = mp \, Isp \, Go$$

Equation 1. Propellant mass ratio with total impulse

Where go is the acceleration of gravity, ISP is the specific impulse, mp is the propellant mass and it is the total impulse, given by:

$$It = \int F dt$$

Equation 2. Total Impulse

With F being the average thrust of the engine, we have a propellant mass of 1,8 kg. Given the density of the propellant, a volume of 0.00098 m³ was found. Knowing that the common volume fraction of combustion chamber propellant is 0.75 [2], the volume and length of the combustion chamber were also defined, being 0.00107 m³ and 0.520 m respectively.

The web fraction is the web thickness ratio of the propellant to its outer radius and is given by the following formula.



$$wf = \frac{D-d}{D} = \frac{2r Tb}{D}$$

Equation 3. Web Fraction

Using previously displayed values of engine burn rate and burn time and assuming a diameter value for the grain of 0.052 m, 0.0005 m smaller than the diameter of the combustion chamber due to the future existence of the inhibitor layer, a web fraction value of 0.5885 was obtained. Using the first equality and the fraction value, we could find the diameter value of the grain hole, 0.01 m.

To control the shape of the thrust curve and thereby obtain the curve determined in the design requirements, it is necessary to control the KN profile, a dimensionless value representing the ratio between the flare surface area and the nozzle throat area cross-section. The following formula allows the calculation of the grain length with that the starting KN and the final KN are equal, thus generating a more consistent thrust curve.

$$Lo = \frac{3D+d_o}{2}$$

Equation 4. Grain length for symmetrical profile

Thus the ideal grain length for a symmetrical profile is 0.0887 m. Then determine the amount of grain required to comply with the data obtained above, it is necessary to calculate the unit volume of the grain.

$$V = \frac{\pi(D_2-d_2)L}{24}$$

Equation 5. Tubular grain volume

Thus the volume of grain is 0.00098 m³, dividing the total propellant volume value by this grain value was the amount of grain, 0.00001225 m³, thus rounding to 8 propellant grains.

2.1.4. Software and Simulation

To perform the burning simulations, we used the SRM_2014 Excel software. The program consists of filling in specific data previously calculated by the team, and from these, the program performs the simulation of the fuel burn, disposing the data and graphic records on the screen. It should be mentioned that the standard nozzle values of the program were used, ie a half convergence angle of 30 ° and half divergence angle of 12 ° and expansion ratio Ae / At 10,89, the value of the cross-sectional area. throat was calculated by the program itself from the KN of the maximum burn area, where the highest pressure is found, its value was then 6.95Mpa. To perform the simulation, it was determined by the team that the inner diameter of the chamber and the inner length of the chamber would be 52.5mm and 520 mm respectively. After a series of simulations changing segment values and some measurements, it was concluded that the configuration that presented better performance compared to the desired propellant (KNSU) configuration was with a 52mm diameter propellant grain with 8 segments of 65mm each. Thus, the engine would achieve a total thrust of 2200N-sec with a maximum thrust of 1008 psi and average thrust of 1641 N. Following are the graphs related to engine performance. According to the data obtained in the simulation and from the calculations performed by the propulsion subsystem, it was decided that the best configuration to achieve the desired performance from the defined dimensions is the previously defined 8-segment configuration, which allows the engine reaches the desired thrust without exceeding the maximum operating pressure of 1000 psi. To perform the simulation, it was determined by the team that the inner diameter of the chamber and the inner length of the chamber would be 54mm and 550mm respectively. After a series of simulations changing segment values and some measurements, it was determined that the configuration that presented the best performance compared to the desired propellant (KNSU) was with a propellant grain.

Following are the graphs related to engine performance.

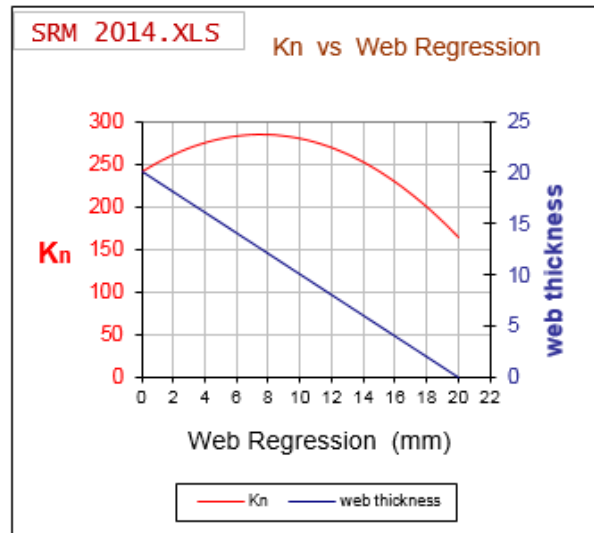


Figura 14 - Regression Chart KNSU

According to the data obtained in the simulation and from the calculations performed by the propulsion subsystem, it was decided that the best configuration to achieve the desired performance from the defined dimensions is the previously defined 8-segment configuration, which allows the engine reaches the desired thrust without exceeding the maximum operating pressure of 1000 psi.

2.1.5. Selecting the Inhibitor Material

To build a rocket engine, certain steps must be taken. One of these is the decision to use a good inhibitor to burn propellant in the combustion chamber. The purpose of the inhibitor is to prevent combustion on a specific surface, ie it is a thermally resistant material that prevents the burning that is already happening in the propellant from reaching the surface of the combustion chamber. This section is intended to show the simulations and data used for choosing an efficient engine inhibitor, taking into account that the inhibitor is an extremely important component during propellant burning.

Simulation and Data Collection For the choice of inhibitor, data collected by Richard Nakka, an American mechanical engineer specializing in rockets, was used. Using a series of inhibitors of different thicknesses, Nakka performed experiments to test the ignition time of each material. The experiment carried out consists of attaching an inhibitor strip of a certain thickness to a propellant tube both above a torch facing perpendicular to the inhibited surface. The propellant pipe was isolated from the torch flame by a metal plate to prevent combustion by the torch.

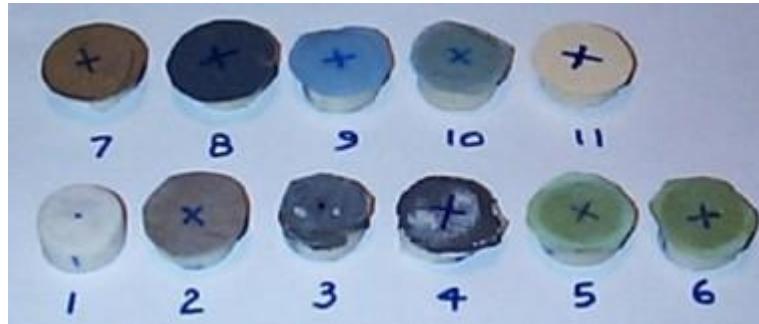


Figura 15 - torched materials

Source: Richard Nakka

For the experiment, the inhibitors and thicknesses present in figure were used.

Sample:	Inhib. Code	Inhibitor material	thickness inhibitor (inch)
1	0	None	0.000
2	CAP	Cardboard, t=.023", painted, bonded with c.c.	0.027
3	SS1	Sodium silicate/cotton fabric (.010")	0.040
4	SS2	Sodium silicate/cotton fabric (.020")	0.050
5	NEO	Neoprene/cotton fabric (.010")	0.011
6	NEO	Neoprene/cotton fabric (.010")	0.013
7	CA	Cardboard, t=.023", plain, bonded with c.c.	0.026
8	PVC	PVC sheet, .033", bonded with c.c.	0.039
9	EPO	Epoxy/cotton fabric (.010")	0.012 (est.)
10	POL	Polyester/cotton fabric (.010")	0.013 (est.)
11	TAG	Tagboard, .011", bonded with c.c.	0.012

Figura 16 - thickness of materials

Source: Richard Nakka

The next step of the experiment is to activate the torch at full power and set a timer for the time taken for the propellant to ignite. From the data it was possible to assemble the table present in figure 3.

Sample:	thickness inhibitor net (inch)	time to ignition (sec.)	"normalized" time to ignition* (sec.)	Inhibitor code
1	0.000	1.3	—	0
2	0.027	9.0	3.4	CAP
3	0.040	4.5	1.1	SS1
4	0.050	12.8	2.6	SS2
5	0.011	3.7	3.4	NEO
6	0.013	5.7	4.5	NEO
7	0.026	12.2	4.7	CA
8	0.039	24.8	6.4	PVC
9	0.012	8.9	7.4	EPO
10	0.013	10.0	7.7	POL
11	0.012	7.1	5.9	TAG
* normalized to .010" inhibitor thickness				

Figura 17 Experiment results table

Source: Richard Nakka

Figure above shows the inhibitors respectively, their thickness, the ignition time, the “normalized” ignition time and the inhibitor code in question. “Normalized” time refers to the time for firing with a fixed thickness of 0.010 inches. To determine this time, equation 1 was used.

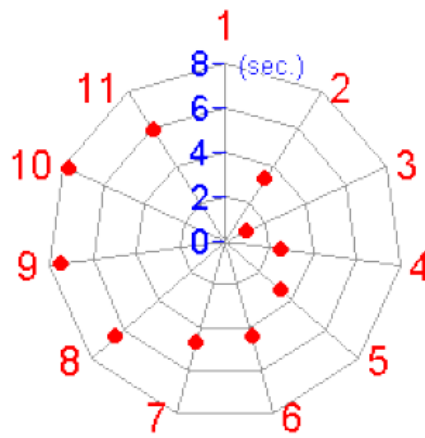
$$Tn = \frac{0,010T_{ign}}{t_{act}}$$

Equation 6- Normalized ignition

Where:

- Tn = Normalized time
- T_{ign} = Burning time at indicated thickness
- t_{act} = Inhibitor Thickness Thus, it is possible to compare the effectiveness of each inhibitor with the same thickness.

From the normalized burn time, it is possible to set up an efficiency graph, which is shown in figure 4, which shows the ignition time of the samples with the listed inhibitors.



Sample: Inhibitor material

- 1 None
- 2 Cardboard, t=.023", painted, bonded with c.c.
- 3 Sodium silicate/cotton fabric (.010")
- 4 Sodium silicate/cotton fabric (.020")
- 5 Neoprene/cotton fabric (.010")
- 6 Neoprene/cotton fabric (.010")
- 7 Cardboard, t=.023", plain, bonded with c.c.
- 8 PVC sheet, .033", bonded with c.c.
- 9 Epoxy/cotton fabric (.010")
- 10 Polyester/cotton fabric (.010")
- 11 Tagboard, .011", bonded with c.c.

Figura 18 - Normalized Ignition Chart

Source: Richard Nakka

From the simulation data, a Pugh Matrix was assembled with the four best inhibitors to compare them with regard to cost and maneuverability.

A Pugh Array is an array in Excel that compares items with a base item. Thus, items are evaluated based on positive or negative scores in relation to the base item, with several criteria with different weights.

PUGH CONCEPT SELECTION

INIBIDOR		DESIGN CONCEPTS			
		baseline image	concept image	concept image	concept image
		PVC	Algodão/ poliester	Algodão/ epóxi	Tagboard (canson)
Criteria/Concept	Weigth	Baseline	Concept 1	Concept 2	Concept 3
Custo	3	0	0	-2	+1
Manuseabilidade	2	0	-1	0	-1
Relação tempo para ignição/espessura	3	0	+2	+2	-1
Total Σ +		0	2	2	1
Total Σ -		0	1	2	2
Total Σ 0		0	0	0	0
Total with weigth		0	4	0	-2
Rank with weigth		2	1	2	4

Figura 19 - PUGH inhibitor

The Pugh matrix for the inhibitor is shown in figure above.

According to the data obtained in the simulation, with the Pugh Matrix and from the calculations performed for the unique efficiency of each inhibitor of the same thickness, the Rocket Team propulsion subsystem - UTFPR / CP

decided to use the cotton inhibitor with polyester resin due to its ease of obtaining, as well as not high cost, and mainly due to the fact that presents the best results during the simulations performed. However, when researching the products, it was decided to replace the polyester with the epoxy resin due to better handling during inhibitor production as well as limitations during the engine machining process. At first, the change will not drastically affect engine performance, so the design will continue with the change.

□ Thermal inhibitor: Due to combustion temperatures (1700°C) above the melting point of steel, the inhibitor aims to create a contact resistance and decrease heat exchange between the interior of the combustion chamber and the outside. It is made of cotton fiber composite and epoxy resin measuring 2mm thick.

2.1.6. Nozzle simulation and design

Preprocessing: Gmsh;

Processing: OpenFOAM - rhoCentralFoam;

Post Processing: ParaView.

Data: Propep 3

In order for the results to be developed in this work to be acceptable, it is necessary to validate the case structure assembled in OpenFOAM. NASA on its website has a general case of convergent-divergent nozzle with analytical results with the explicit purpose validate CFD codes by comparing with the analytical data provided.

This case is composed of three different flows where the inlet pressure in the nozzle is fixed and outlet pressure is varied, the other conditions proposed. The geometry of this nozzle is generated with a small algorithm provided by NASA that creates values for the area according to the distance x in the nozzle.

This way we can validate the results of this simulations

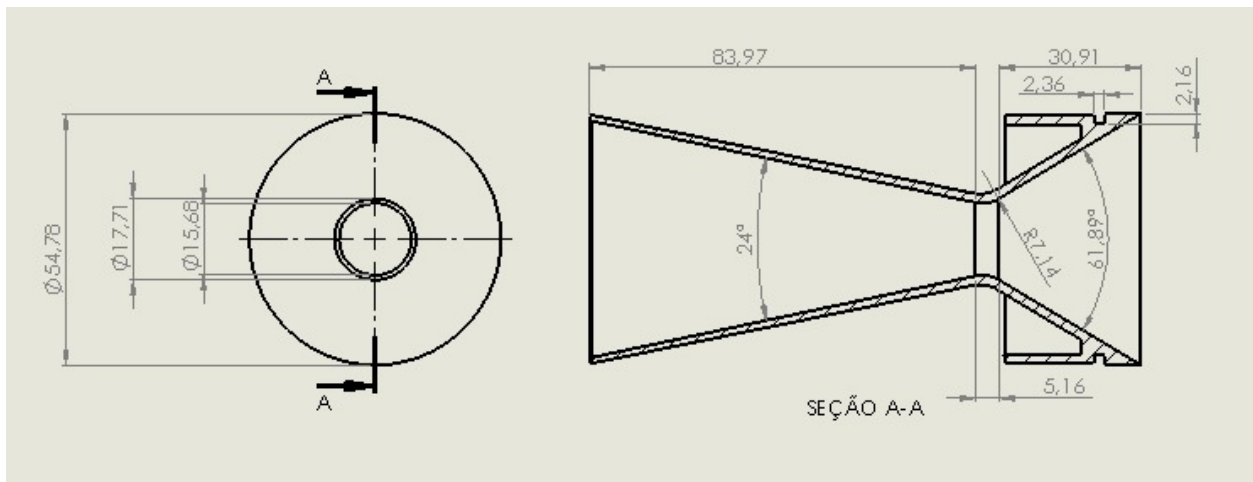


Figura 20 - Nozzle project

2.1.7. INITIAL AND CONTOUR CONDITIONS

In the constant folder, OpenFOAM has a file called boundary, where conditions basic contour lines should be described, these conditions were defined from the regions generated by gmsh. In this way the two faces of the cylinder were defined with the "wedge" type, the nozzle wall was defined as the "wall" type, in Portuguese, nozzle inlet and outlet were "patch" type, a general type for like these.



For pressure conditions the "TotalPressure" function was used for the nozzle inlet, This condition is ideal for places where pressure is known but not speed, as in this case convergent-divergent nozzles. In all three flows a value was used here.

10000 Pa. It was also decided that the working fluid would be air, thus defining the Specific heat ratio k as 1.4.

28

In the nozzle wall the condition used was "ZeroGradient", this condition goes beyond a value for the contour from the nearest cell. For the nozzle outlet was then applied to waveTransmissive condition this condition ensures that no waves will be reflected back to the nozzle and determines the external pressure of the case, for simulated flows these pressures 8900 Pa, 7500 Pa and 1600 Pa, thus ensuring the proposed pressure ratio of 0.89, 0.75 and 0.16.

In case of temperature, the nozzle input has the condition "fixedValue" establishing a fixed value of 298 Kelvin, thus representing the combustion chamber of the system. The conditions wall and nozzle outlet are "zeroGradient" thus allowing the simulation itself find the values for these regions from the given values.

In the case of speed, the field entered was (0 0 0), no speed in the system, the conditions applied to the input and output was "zeroGradient", in the "Slip" condition is used since the boundary layer on the wall is not of interest in this analyze.

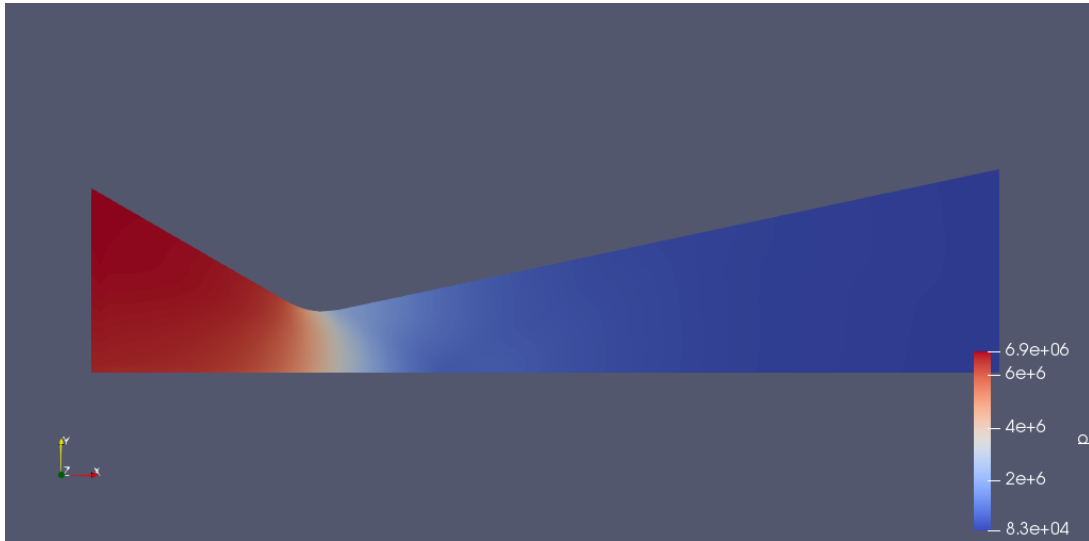


Figura 21 - Pressure profile

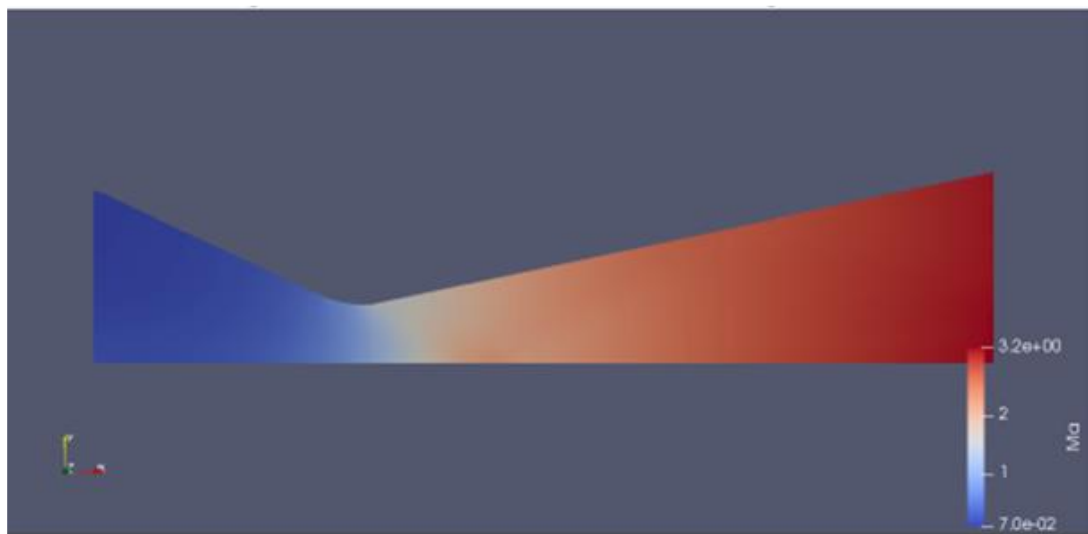


Figura 22 - nozzle speed profile

Assumptions about runoff:

- Quasi-dimensional;
- Compressible;
- Ideal gas;
- Isoentropic;
- Permanent regime.

Applying the conservation equations to the nozzle and using the hypotheses discussed, the following equation is obtained.

$$\frac{dA}{A} = (Ma^2 - 1) \frac{du}{u}$$

Equation 7 – conservation of energy

After a number of changes we concluded that a 7mm radius on the throat is needed to prevent backflow in the divergent section.

The nozzle geometry was optimized through flow simulations, seeking a perfectly expanded exhaust and higher exhaust velocity, reaching Mach 3.2 (3951.36 km / h). As with the other structural elements, simulations have been made to certify safety with the lowest possible weight.

2.2. Avionics

2.2.1. DEVELOPMENT

Considering the shape of the rocket where the avionic system will be positioned we opted for a modular system. The system will consist of several printed circuit-shaped discs with specific functions stack on top of each other and can add, change or remove with ease when needed. Thus, enabling a system upgrade only adding a plate.

For the Rocket UTFPR CP rocket, 4 layers were considered they are the load and power control system, onboard computer system, telemetry system and telecommunication system.

2.2.2. LOAD AND POWER CONTROL SYSTEM

The power load control system is responsible for providing power to the other systems. It consists of battery, voltage regulators and protection systems.

For the development of the feeding system, the other systems to know the general and specific power needs of each.

From this, the necessary voltage levels were defined, which were 5V and 3.3V. So it was defined that two voltage regulators would be used to satisfy that need. A consumption survey was made for each of the sectors .

Tabela 2 - Energy consumption

Setor	Consumo(mA)	
	3,3V	5V
Computador de bordo	400	
Telemetria	5,5	
Telecomunicação	650	
Recuperação		10
Consumo total	1055,5	10

Then the power outputs for each converter were calculated in order to make the best choice of each as well as the battery. As it shows the diagram of Figure 9.

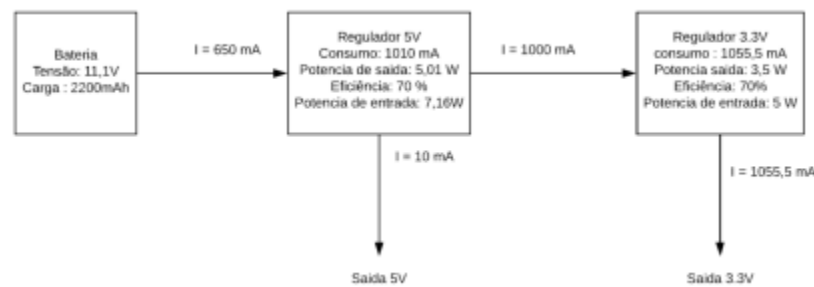


Figura 23 - Block Diagram

Tabela 3 - Consumption of sectors by supply voltage

	Componente	Tensão de Entrada	Tensão de Saída	Potência Máxima Indicada	Tensão Configurada na Saída
U1	Regulador de Tensão LM2596	3,2V-40V	1,5V-35V	15W	5V
U2	Regulador de Tensão LM2596	3,2V-40V	1,5V-35V	15W	3,3V

	Componente	Tensão de Saída	Carga Máxima	Dimensões (mm)	Peso (g)
BT1	Bateria Lipo 11.1V 2200mAh	11.1V	2200mAh	25x35x107	172

With the parameters defined, the available components were chosen in the market, and its settings are defined for the proper functioning of the system. As shown in Table 3.

At first there were two options to choose from, either Lithium or Li-Po, however, the Li-Po battery was chosen because it is more malleable, has a lower dilation with respect to heating and has a higher energy density making it smaller and lighter, resulting in better design efficiency.

For sizing the battery the following data were used provided by each sector.

The recovery sector contains the servo which has a peak current of 1.2A, but it is for a Whimsy time. This fact was considered only in the maximum drainage of the battery. By being fed at 5V this 1.2 A peak generates a 500mA peak in the battery. From them, the current was calculated drained into the battery.

With the output current in the battery, the charge required for satisfy the 3 hours requested in the competition. As shown below.

Cosumo máximo da bateria (mA)	650
Tempo desejado (Horas)	3
Carga nescessaria (mAh)	1950

The Lipo type battery proved to be the best of options because it provides close to the desired load, avoiding oversizing in addition to its physical characteristics of size and mass that were smaller than the other options listed. Another very important feature is the high ability to supply current instantaneous.

After the circuit design with the converters, the circuit board layout was circuit board, shown below.

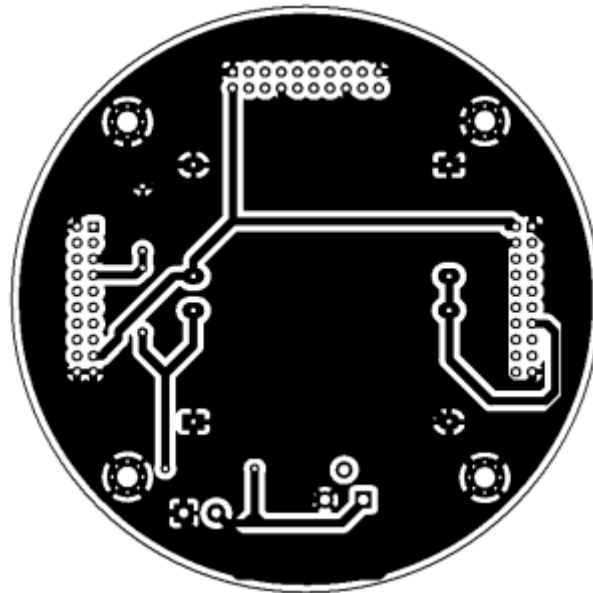


Figura 24 - Load Control System

2.2.3. BOARD COMPUTER SYSTEM

The on-board computer system shall be responsible for processing data and control of the entire avionics system and a datalogger that will perform the storage of all data obtained during the flight for future analysis and studies.

To achieve the objectives of the on-board computer system, a processing center and a device for storing data. The ESP32 microcontroller was chosen for data processing and system control.

from Espressif Systems, for its simplicity of programming, low cost relative to their processing potential and ease of purchase (purchase).

For data storage we chose a micro SD card for its size reduced weight and ease of use. The access to the micro SD will be made from a micro SD recorder module.

Programming of the ESP32 microcontroller will be performed using the available from Espressif Systems, called ESP-IDF (Espressif IoT Development Framework) in C language.

The on-board computer will be responsible for reading the system sensors.

telemetry and their storage and processing in order to perform the launch start and end detection to maximize system autonomy. Beyond In addition, it will be responsible for the performance of the parachute at the exact moment required by along with the real-time transmission of all data during and after flight in order to perform vehicle recovery.

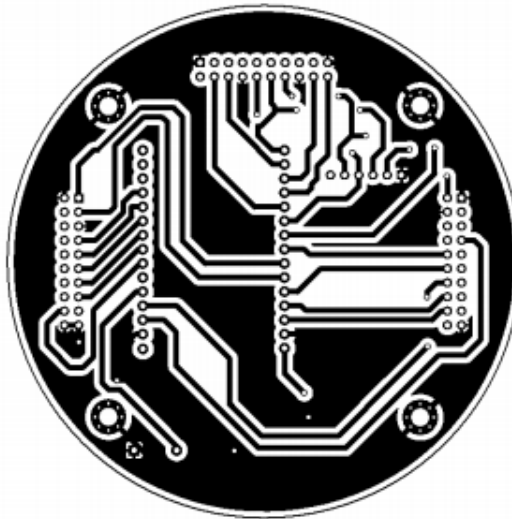


Figura 25 - Board computer system

2.2.4. TELEMETRY SYSTEM

The function of the telemetry system is to be responsible for tracking the rocket, parachute actuation and flight data such as acceleration, temperature and pressure. For such operation of the telemetry system components with gyro sensors, accelerometer, barometric and temperature.

The 9250 MPU is a chip that has a 3-axis accelerometer, a gyroscope 3-axis and a 3-axis magnetometer (magnetic field sensor).

The HMC5883L module contains a 3-axis magnetic sensor that provides X, Y, and Z axis information with 1 to 2 degree accuracy automatic calibration.

Type K Thermocouple with MAX6675 module contains a temperature sensor which handles incoming signals accurately from -20 to + 85 ° C, its margin of error is at ± 5 ° C.

The BMP 280 module performs pressure measurements to within ± 1 hPa and temperature accurate to within ± 1 ° C. With this accuracy, it is possible to perform measurements altitude with a margin of error of ± 1 meter. With regard to pressure measurement, the BMP 280 reads between 300 and 1100 hPa, which corresponds to the altitude range from - 500 to + 9000 m.

The main reasons for choosing these devices was that there were the need to capture all data efficiently, withstanding all the pressure and physical situations that could occur internally, in addition to considering the high speed the rocket reaches. So as it is a delicate project the simpler components that did not support such precision requirements were disregarded; the preference was for domestic products which could facilitate the whole process of purchase. Finally, four modules for the project ultimately aiming at both cost and benefit

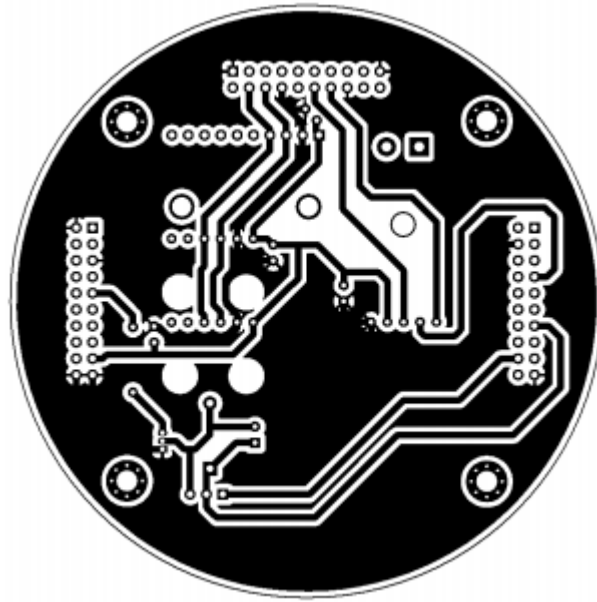


Figura 26 - Telemetry System

2.2.5. TELECOMMUNICATION SYSTEM

The Telecommunications sector is responsible for data transcription and the obtaining the geolocation. For this project we used the communication module LoRa E32-TTL100 1W and the NEO6M GPS geolocation module.

LoRa network is a wireless solution in networks below GHz, in frequencies that do not require licensing such as 433, 868 and 915 MHz. It is a type of network being used to connect devices to low-end applications consumption, long distance, since in the open field it is possible to reach up to 15 km of range. Simply put, the Lora system is a module placed in the devices, as in this project's station, and also in gateways, which can get data out of a LoRa network and through a IP connection reach local and / or remote servers. Modules send and receive gateways, similarly to Wi-Fi networks.

LoRa has advantages compared to other types of communication, such as mobile phone, Bluetooth Low Energy (BLE), or WiFi. Suitable for IoT systems such as sensors and monitors especially those that are battery-powered and coverage area.

The LoRa module has its detailed specifications in its datasheet, the Information that is most important to the project at the moment is listed below:

- Operating temperature: -40°C to 85°C;
- Operating humidity: 10% to 90%;
- Transmission Current: Minimum = 570mA, Typical = 610mA, Maximum = 670mA
Up to 30% above current supply range recommended nominal value, ie a source with a capacity of 800mA;
- Receive Current: Minimum = 19mA, Typical = 20mA, Maximum = 22mA
- Sleep Mode Current: Current required for when the module is in Sleep / Very Low Power operating mode - Minimum = 4uA, Typical = 5uA, Maximum = 6uA;
- Supply voltage: Minimum = 2.3V, Typical = 3.3V, Maximum = 5.2V
- Transmission power 30.0 dBm;
- Reception sensitivity -147.0 dBm.

The NEO-6M GPS shield module is extremely easy to use by only have 4 connection pins (GND, VCC, RX and TX) and communication standard serial. Still it is a high performance module for use with microcontrollers, where it is compatible with Arduino and Raspberry PI. This module is able to report exact location by sending latitude and longitude, date, time and travel speed through 3.3 serial interface V.

The NEO-6M GPS is used for the acquisition of coordinates for the below, some information about it, which may be found in detail in your datasheet.

- Supply voltage: 3.3V to 5.0V (module power only, communication needs to be up to 3.3V since the module has a voltage regulator);
- Communication Type: UART TTL 3.3V;
- Communication Baud Rate: 9600 (Standard);
- Communication voltage: Maximum input voltage for low logic level: 0.6V Minimum input voltage for high logic level: 2.31V Maximum output voltage for low logic level: 0.4V Maximum output voltage for high logic level: 2.9V
- Dimensions of the module (WxHxH): 30x23x4mm;
- Dimensions of the antenna (WxHxH): 25x25x8mm;
- Start times:

Time required for GPS to receive data,

Cold start starting the whole system without power, and Hot Start using a battery

Backup Cold Start: 27s Hot Start: 1s

- Maximum refresh rate: 5Hz • Horizontal position accuracy: 2.5m
- Operating limits: Dynamic: $\leq 4g$, Altitude: 50000m, Speed: 500m / s

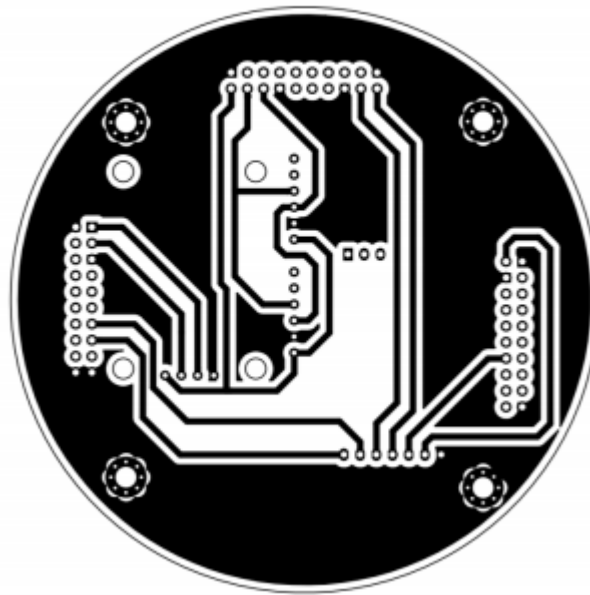


Figura 27 - Telecom System

2.3. Structural



2.3.1. Fins and cone Material

From the Pugh matrix, considering our decision parameters, we chose the ABS (Acrylonitrile Butadiene Styrene) material as it is a low cost material and be easy to manufacture and accessible, a priority for our manufacturing reality, as we show from Pugh's matrix.

Decision Parameters

We use the Pugh table method which consists in assigning a rank of our pre-established criteria and the concepts of each material according to each criterion.

The criterion is ranked from 1 to 5, with 1 being the least relevant and 5 the most relevant according to our manufacturing process and location. As well as the materials, which for each criterion earn a rating from -2 to +2, with -2 being the lowest performing material in given criterion, +2 the highest performance and 0 the base material.

The decision is made by multiplying the value of the criteria by the values of the ratings of the this gives a weighted sum that determines which material is the best to choose according to our needs.

It was agreed between the members of the structure subsystem, and from the method that the best material for couplers is nylon6 due to its weight, cost and usability are the best among the materials studied.

From the materials presented earlier in the report, it was suggested to the group of the framework subsystem that a PUGH array was assembled with the defined criteria for all together. This way you can make a more detailed analysis of the material to be to choose, taking into consideration which criteria should have the greatest or least influence depending on the project proposal in question.

The application of fiber has the objective of promoting a significant increase in the resistances parts of ABS printed parts such as bending and impact resistance. The resin was used by ease and wide range of applications, as well as providing a finer surface finish, compared to ABS. Further studies may qualify and quantify such improvements.

Constructive Description:

• Fins:

1. Modeling:

The. Once geometry is defined, model the fins in 3D modeling software and save them in the format required for 3D printing - usually STL;

B. Pay attention to the print parameters - fill type, print type workmanship, material, color, etc. - must be in accordance with the project;

2. Preparation

The. After receiving the items, prepare the surface of the fins with medium sandpaper make them rough enough for the epoxy resin and fiber to adhere of glass;

B. Use the fins as a mold to cut the sized fiberglass fabric slightly larger than the fins, one flap for each face of each fin;

W. Prepare epoxy mixture - 2/3 resin and 1/3 hardener - in quantity only sufficient for application on one side of all fins;

3. Application:

The. Apply, by spatula or similar object, the mixture on one side of each fin;

B. Laminate the cut fabric over the applied resin, checking for any air bubble or non-compliance, remove them if any;

W. Apply another layer of resin on the fabric, creating small resin blade;

d. Attention to the resin working time, the application steps should be done

As soon as possible, the resin begins to harden after approximately 18 minutes after mixing; In the case of many pieces for application, divide them into portions in order to work with small doses of mixture;

and. Separate parts in place to wait for partial cure time for approximately 5 hours;

f. After partial cure, repeat the application steps for the other side of the fins;

g. Wait for total resin cure time of 5 days;

4. Finishing:

The. Remove excess edges with stylus;
B. Smooth the edges of the fins with sandpaper;
W. Using a precision balance, measure the masses of the fins and record them; use the piece with lower mass as a base parameter;
d. Using the angle sander, remove material from the excess grease fins until stay balanced;
and. Perform the final finish with sandpaper.

2.3.2. Centralizers

The greater weight (3) is for the specific mass of the component, a lighter material will improve design performance. Cost weighed (3) because one of the team's current needs is to spend the least amount of money per application on each project. Safety, ease of manufacture, are weighted (1) because from special measures that can be taken such as use of PPE, people able to handle the most dangerous equipment, all tend to reduce the risks and difficulties. involved in the manufacture of these parts. The melting temperature has weight (1) as it will have no practical effect on design. The strength of the material received weight (2) because we need a material that can withstand the stresses provided by the project. Criterion Wood Aluminum Nylon 6 Density 590 kg / m³ 2,71g / cm³ 1,14g / cm³ Yield limit 13 Mpa 12,7 Mpa 60 Mpa Tensile strength 3 Mpa 48 Mpa 50 Mpa Tensile modulus 8846 Mpa 7000 Mpa 190-210 Gpa

2.3.3. Frame

We use the Pugh table method which consists in assigning a rank of our pre-established criteria and the concepts of each material according to each criterion.

The criterion is ranked from 1 to 5, with 1 being the least relevant and 5 the most relevant according to our manufacturing process and location. As well as the materials, which for each criterion earn a rating from -2 to +2, with -2 being the lowest performing material in given criterion, +2 the highest performance and 0 the base material.

The decision is made by multiplying the value of the criteria by the values of the ratings. This gives a weighted sum that determines which material is the best to choose according to our needs.

Cost ranked highest 5 due to minimum disbursement of money per application on each project and to save resources for unforeseen during its realization. Accessibility is ranked 4 because it is notorious importance the easy access of the material in the market and its purchase request through the Institution. Rank 3 is the ease of manufacture because the easy handling of the material streamlines the project by providing more time for resolving contingencies and of tests. The mechanical strength of the material, although very important, is ranked 2 due to the suitability of the team objective. Both materials have strength sufficient for the purpose they will have in the project. Lastly, in rank 1, was the density.

Preferring lower density materials as it would add less mass to the project and facilitate the recovery of the rocket on the high seas.

Some technical parameters of the decision refer to the table below:

Tabela 4 - Materials table

Cr�terio	PVC	Fibra de Carbone	Fenolite
Densidade	1,30 g/cm ³	1,76 g/cm ³	1,35 g/cm ³
Resist�ncia � tra��o	50,33 MPa	16 MPa	135 MPa
Resist�ncia � compreens�o	49,99 MPa	126 MPa	200 MPa

It was agreed between the members of the structural subsystem, and from the method of evaluation adopted, that the best material for the current condition of the team is PVC. The sizing and calculations related to structure and finite elements will be presented in other reports.

2.4. Recovery System

Project Parameters

Firstly, a parachute type parachute was chosen as it provides greater strength in the tie rods and less material is required in its inflated shape.

Regarding the shape of the parachute, we chose a semi-ellipsoidal parachute with 12 buds, in which the cross-sectional shape is semi-elliptical. The advantage of this format is that less material will be used to make the parachute, thus providing less weight.

For the parachute sizing (Figure 1), it was adopted that the ratio of the semihelipse, ie the ratio between height (b) and radius of the base (a) should be 0.707, a value that, according to Nakka, provides the distribution. more favorable canopy tension and provide a better drag coefficient.

However, the determination of such dimensions depends on some parameters, such as: rocket terminal velocity (v), rocket mass (m), drag constant (k), and air density at the launch site. The expression relating these terms to the Parachute Useful Area (A) is shown in Equation below.

$$A = \frac{2.m.g}{\rho.k.V^2}$$

Equation 8 – area of chute for velocity

Adopting $v = 8 \text{ m/s}$, this speed will not compromise rocket recovery as it will fall into the ocean, allowing us a greater safety margin, $m = 6.5 \text{ kg}$, $k = 0.77$ (for the specified parachute type) and $\rho = 1,229 \text{ kg/m}^3$ we have the Parachute Useful Area (A) is 2,105 m^2 .

Thus, from the data found, we obtain the radius of the mushroom base (a) and the height of the mushroom in relation to the base (b), exemplified in

Tabela 5 - chute dimensions

Área Útil (m^2)	a (m)	b (m)
2,105	0,688	0,486

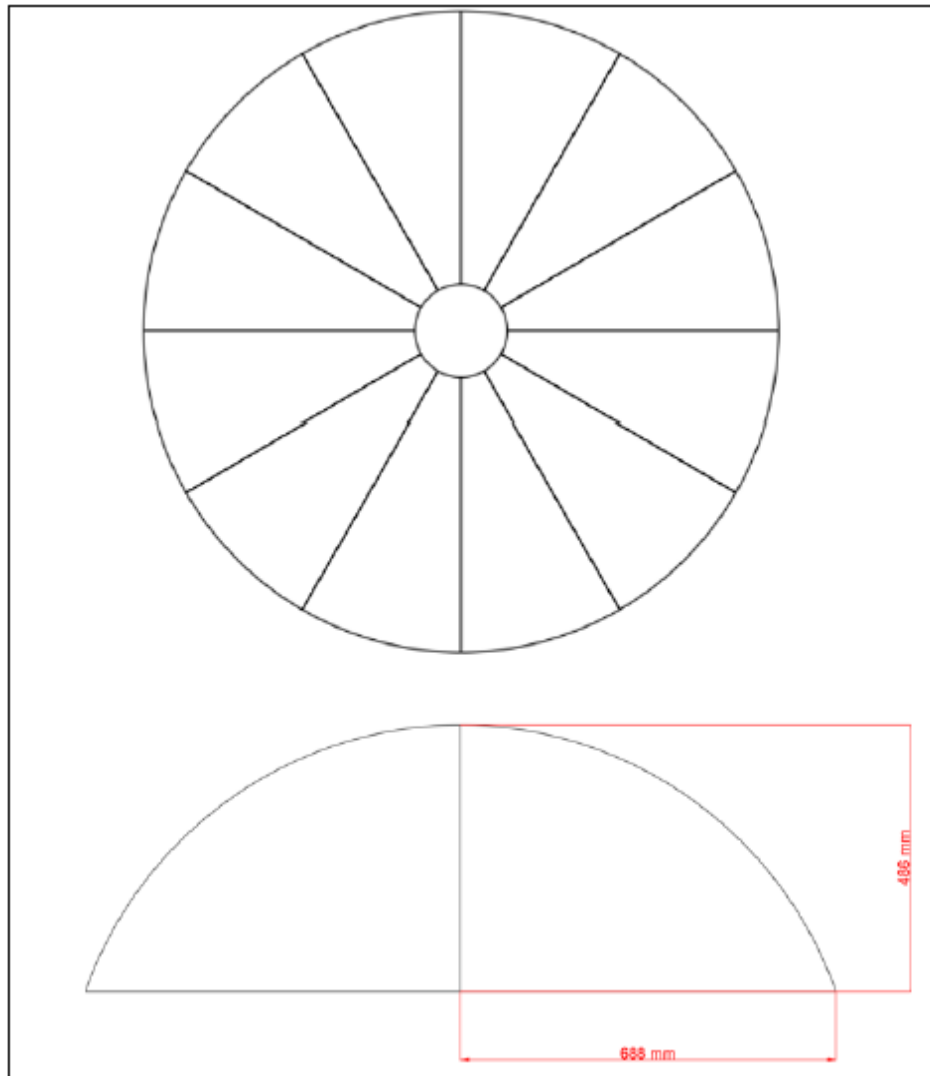


Figura 28 - Semi-episoidal chute

It was also possible to estimate the tissue mass used to make the parachute, from the Mushroom Base Area ($A_b = 1.48 \text{ m}^2$) and tissue density ($d = 1.55 \text{ kg} / \text{m}^2$). Therefore, the estimated mass was 0.326 kg .

Finally, for the parachute 12 buds were chosen, as recommended by Nakka. The shape of the buds was calculated so that the inflated parachute took the form of a semi-ellipsoidal shell, respecting the height-radius ratio.

3. Theoretical Validation Results

3.1. Propulsion Results

3.1.1. Engine information

Grain total mass : 1759kg
 Grain density : 1,795 kg/cm³
 Number of grains : 8
 Grain mass : 219,87 g
 Grain length: 65mm
 Space between grains: 3,75 mm
 Core rod diameter : 10mm
 Core rod model: BATES
 Target pressure: 1000 psi
 Chamber length inside: 520mm
 Chamber total length : 580mm
 Expansion rate : 10,869

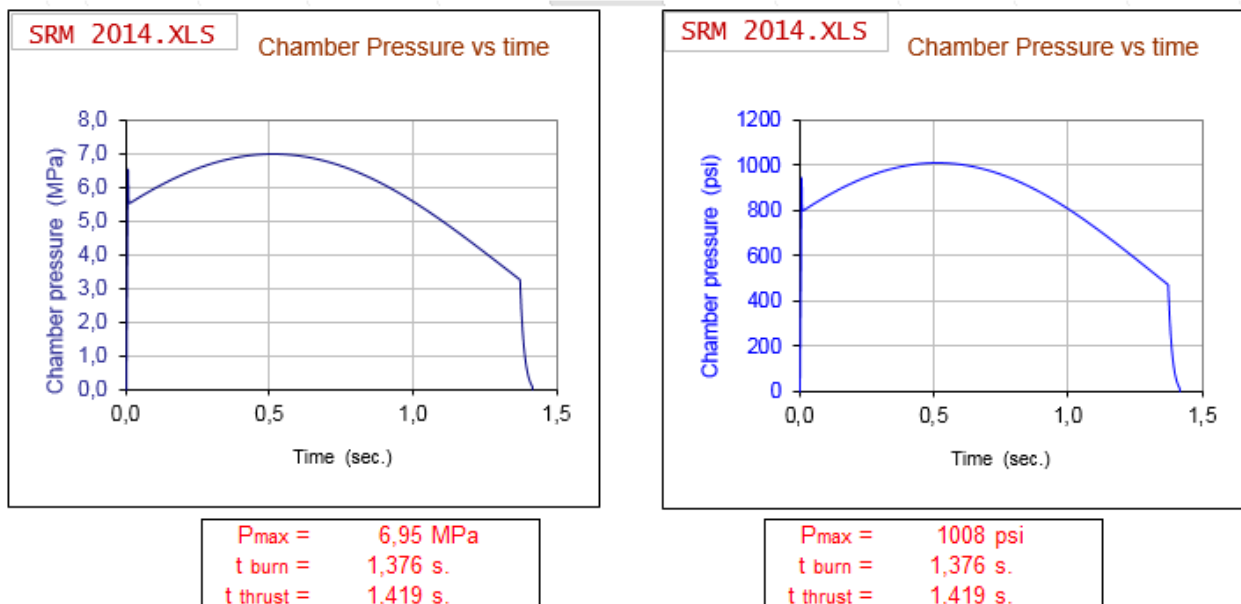
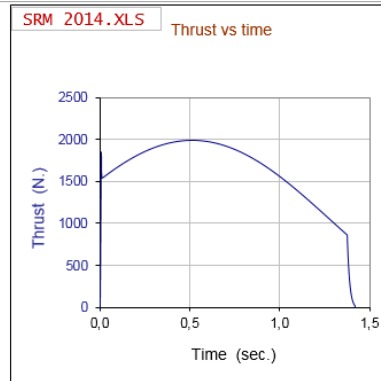


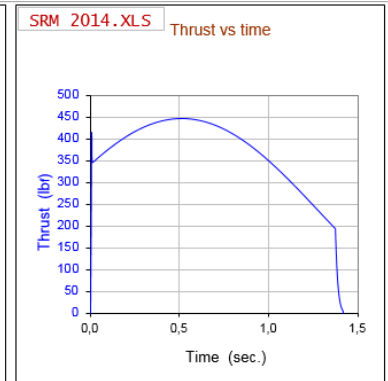
Figura 29 - chamber pressure

Rocket Motor Performance

k	1,044	Ratio of specific heats, 2-ph. flow
η noz	0,85	Nozzle efficiency
Ae/At	11,00	Nozzle expansion ratio (initial)
Patm	0,101 MPa	Ambient pressure
At0	193,1 mm ²	Throat cross-section area (initial)
Ae	2124,4 mm ²	Nozzle exit cross-section area
Me0	3,166 0,00	Mach No. at nozzle exit (initial)
MeF	3,166 0,00	Mach No. at nozzle exit (final)
De	52,01 mm	Nozzle exit diameter
Ae/At opt	12,900	Optimum Nozzle expansion ratio at Po max
Ae/At opt	10,869	Avg. optimum nozzle expansion ratio
w f	0,804	Web fraction
Cfmax	1,461	Thrust coefficient, maximum
F max	1989 N.	Maximum thrust
I t	2300 N-sec.	Total impulse
Isp	133,4 sec.	Specific impulse, delivered
Class:	K	Motor classification



F max = 1989 N.
F avg = 1621 N.
t thrust = 1,419 sec.



F max = 447 lbf
F avg = 364 lbf
t thrust = 1,419 sec.

Figura 30 Thrust curve

We simulated using ansys static structural and transient structural to Project the chamber wall, bulkhead and nozzle wall.

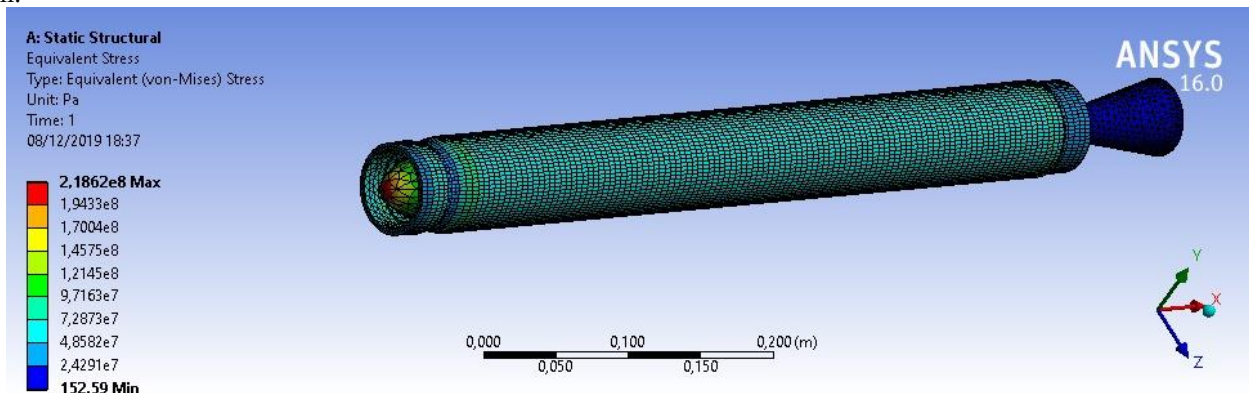


Figura 31 - Engine Max stress

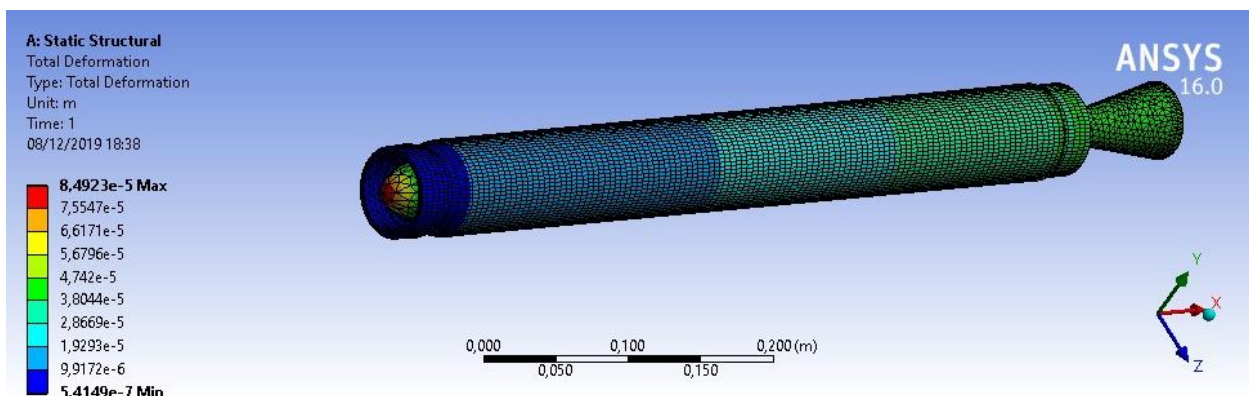


Figura 32 - Engine max deformation

Simulation Data:

- Max stress on the chamber: 97 Mpa



- Max stress on the bulkhead: 218 Mpa
- Max stress on the nozzle: 121 Mpa
- Max deformation: 0,0849mm

3.1.2. Nozzle

Convergent-divergent nozzles are relatively simple devices, an area duct variable, however several factors are responsible for its performance, the inlet pressure and output, the local atmospheric pressure and of course the various possibilities of geometry that end up influencing the performance of the device in one way or another.

Through the computational fluid dynamics it is possible to analyze the flow of these predict shock waves, assess their properties during flow and seek ways to optimize the project. Without the need to perform static tests with the system of propulsion, which carries costs and risks.

From the results obtained in this work it is possible to conclude that certain decisions will positively contribute to system performance by reducing the convergence angle carries an improvement in flow velocity to compensate for the increase in nozzle length it is reasonable to increase the angle of the convergence section, its length can be reduced without noticeable loss of speed.

The nozzle radius cannot be overlooked, although slight variations in size do not appear to have much influence, lack of lightning or possibly a small radius too much may lead to the formation of oblique shock waves in the mouthpiece, such discontinuities are undesirable from the point of view of flow.

The OpenFOAM program proved more than sufficient for the needs presented and the code developed in it and presented in this work can be used with reliable for simulating supersonic flow in small nozzles.

3.2. Avionics

3.2.1. CARD MANUFACTURING PROCESS

To produce the circuit boards of each system it was necessary to print the circuit layout on transfer paper and by thermal method, transfer it to a phenolite board. After the plate has cooled, the paper has been removed and carefully soaked in an iron perchloride solution producing the following reaction: $\text{FeCl}_3 (Aq) + \text{Cu} (S) \rightarrow \text{FeCl}_2 (Aq) + \text{CuCl} (Aq)$, which was accelerated by constantly exposing the surface to more iron perchloride using a brush. With all unmarked copper already removed, the paint was removed using a steel wool. Then, with the copper circuit already printed, it was possible to solder the components of each system on their respective plates. Figures below illustrate a bit of the process.

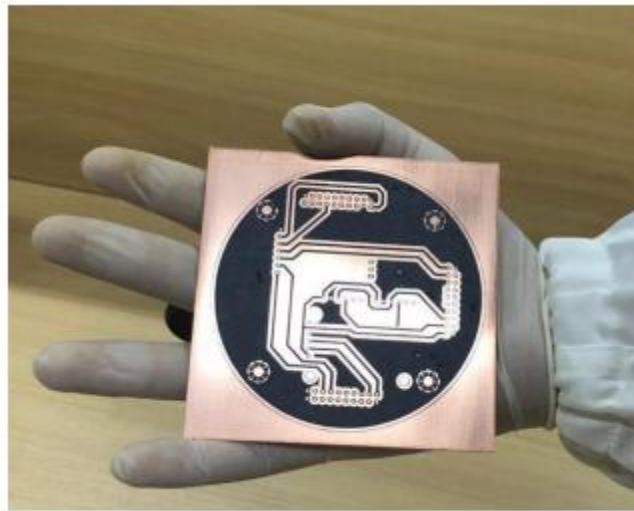


Figura 33 - Circuit printing



Figura 34 - Plate being immersed in iron perchloride



Figura 35 - Method used to accelerate the reaction

Based on the procedures and data presented in this report, it has been possible to develop the avionic project of the team. Once the manufacture is finished the necessary tests to verify the requirements of project. In short, the avionics subsystem will be able to acquire data, process them, store them and transmit them, and act on the recovery system through the servo motor.

3.3. Structures and flight

We simulated flight via open rocket to evaluate the best angle and the critical velocity of the rocket. Using the parameters previously defined we reached a theoretical apogee of 1958m with a maximum acceleration of 29 G and a maximum velocity of 356m/s. we expect a time of 16,9s to apogee.

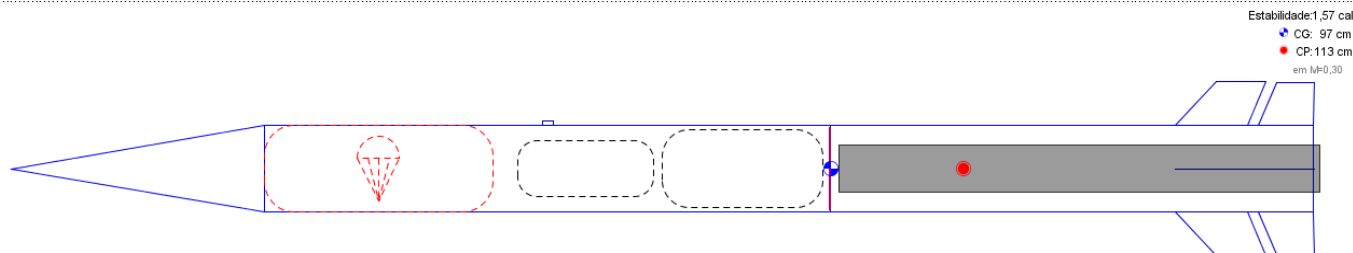


Figura 36 - Structure on Openrocket

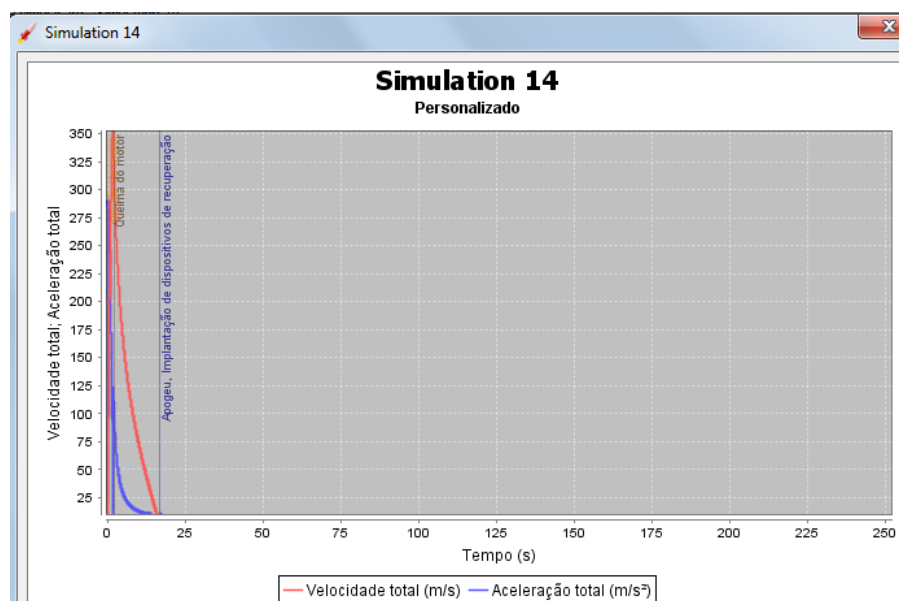


Figura 37 - Maximum Velocity and acceleration

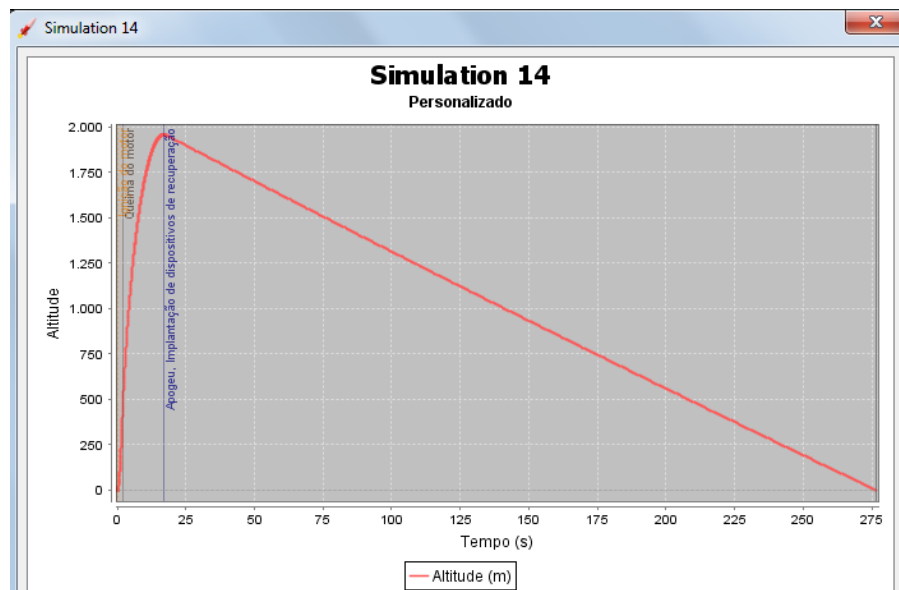


Figura 38 - Altitude Graph

3.4. Recovery System

The recovery system was calculated and simulated via open rocket, using the parameters previously defined we found a recovery speed of 7,21 m/s

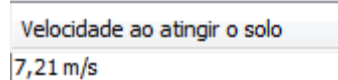


Figura 39 - Terminal velocity

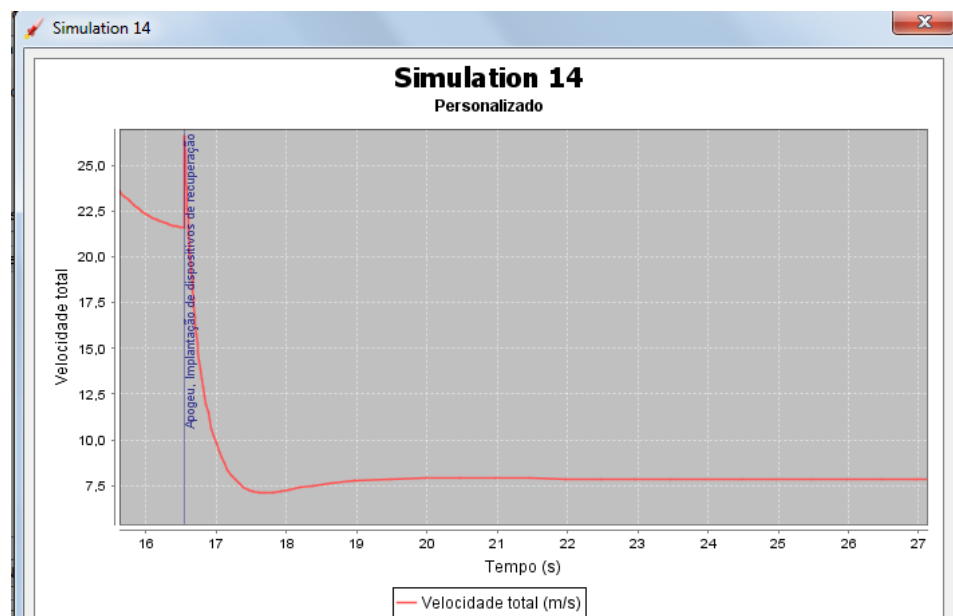


Figura 40 - Terminal velocity Graph

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5. Acknowledgements

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